

Lab 3: Uncertainty in Fluid Flow Measurements

MAAE 2300 L4 - Timeslot 2

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Abstract:

In this experiment various methods of error calculations are used in order to find the uncertainties of different flow meters. In the lab it will be determined which flow measuring devices are the most accurate, taking into consideration the flow conditions, instrumental and measurement uncertainty.

Table 1: Important Results Found

	Flow Rate 1 (m/s)*10 ⁻⁴	Flow Rate 2 (m/s)*10 ⁻⁴
Venturi	1.632 ± 0.0491	1.524 ± 0.0459
Orifice	1.313 ± 0.0135	1.274 ± 0.0131
Rotameter	1.667 ± 0.0833	1.583 ± 0.0792
Turbine	1.669 ± 0.0501	1.546 ± 0.0465
Ultrasonic	1.606 ± 0.0246	1.474 ± 0.0222
Bucket	1.673	1.568

Nomenclature:

g = acceleration due to gravity = 9.81 m/s²

ρ_{water} = density of water = 1000 kg/m³

P = Pressure (Pa)

Q = Volumetric flow rate (m³/s)

C_d = Discharge coefficient = 0.96 for Venturi, 0.65 for Orifice

t = time (s)

D = diameter (m)

d = diameter ratio

u = uncertainty

Procedure:

The procedure for this lab was followed, without change, from Experiment 3 in the lab manual ("MAAE 2300 Fluid Mechanics 1, LABORATORY EXERCISES", Miller, 2023) [1]

Flow Analysis:

For calculating total uncertainty for every flow measuring device, $U_{Total, q}$, all associated errors must be calculated before. The measurement $U_{m, q}$, and the instrument $U_{i, m}$, are combined using the following equation.

$$U_{Total, q} = (U_{m, q}^2 + U_{i, m}^2)^{1/2} \quad (1)$$

Because the Venturi and Orifice meters both measure their respective pressures, the associated error is in units of pressure, so is the associated uncertainty. These error calculations use both the mean and standard deviation formulas.

$$\text{mean} = (\sum \text{observations}) / (\# \text{ of observations}) \quad (2)$$

$$s = (\sum (x - \bar{x})^2 / n)^{1/2} \quad (3)$$

$U_{m, P}$ is calculated first, t_{95} is found using the t-tables at a 95% confidence interval and s is the standard deviation. $U_{i, m}$ is calculated by multiplying the error percentage given by the manufacturer, with the mean of ΔP .

$$U_{m, P} = t_{95} * s \quad (4)$$

$$U_{i, m} = (\% \text{ error}) * (\text{mean}) \quad (5)$$

Using q_{Actual} the flow rate for the orifice meter and venturi is calculated, then derived as a function of ΔP and diameter from Bernoulli's equation. In this equation C_d is the discharge coefficient, D_1 and D_2 are the throat diameters, and d is the diameter ratio between them. The uncertainty of q is composed of ΔP and $U_{m, q}$. Those being the partial derivative of q with respect to ΔP and $U_{m, q}$ is found using equation 8.

$$q_{Actual} = (C_d * D_2^2 * (2(\Delta P) / (p(1 - d^4))))^{1/2} \quad (6)$$

$$\delta q / \delta \Delta P = (\pi * C_d * D_2^2) / (4(2)^{1/2} (p)^{1/2} (\Delta P)^{1/2} (1 - d^4)) \quad (7)$$

$$U_{m, q} = ((\delta q / \delta \Delta P) * (U_{Total, P}))^{1/2} \quad (8)$$

Flow meter uncertainties for Rotameter, Turbine and Ultrasonic sensors are measured. Because of this, equation 1 can be used to combine these individual uncertainties ($U_{i, q}$ and $U_{m, q}$) for the final calculation. 2.2 is still the value used for t_{95} is again 2.2 since it is still a 95% confidence interval. The flow rate measured independently by the bucket can also be calculated. The collected volume (measured in milliliters) is converted to cubic meters then the volumetric flow rate can be calculated with the following equation.

$$q_{Measured} = V / t \quad (9)$$

Analysis and Discussion:

All sample calculations and equations can be found in Appendix B.

A) Tables 2 and 3 show the calculated results for each flowmeter with their respective flow rate, mean and standard deviation.

Table 2: Calculated q Values for Flow 1

	Measured q (m ³ /s)				
Point #	Venturi	Orifice	Rotameter	Turbine	Ultrasonic
1	0.00016364	0.00013261	0.00016666	0.00016716	0.00016156
2	0.00016072	0.00013410	0.00016666	0.00016716	0.00016124
3	0.00016277	0.00013231	0.00016666	0.00016716	0.00016054
4	0.00016479	0.0001308	0.00016666	0.00016716	0.00016064
5	0.00016335	0.0001344	0.00016666	0.00016683	0.00016079
6	0.00016508	0.0001344	0.00016666	0.00016683	0.00016034
7	0.00016277	0.00012552	0.00016666	0.00016683	0.00016043
8	0.00016277	0.00012741	0.00016666	0.00016666	0.00016033
9	0.00016306	0.0001308	0.00016666	0.00016683	0.00015971
10	0.00016335	0.0001308	0.00016666	0.00016683	0.00016053
Mean	0.00016323	0.00013132	0.00016666	0.00016694	0.00016061
Std dev.	1.2021E-06	2.9646E-06	2.8571E-20	1.9324E-07	5.1039E-07

Table 3: Calculated q Values for Flow 2

	Measured q (m ³ /s)				
Point #	Venturi	Orifice	Rotameter	Turbine	Ultrasonic
1	0.00015442	0.00012584	0.00015833	0.00015466	0.00014758
2	0.00014752	0.00012741	0.00015833	0.00015516	0.00014754
3	0.00015257	0.00013261	0.00015833	0.00015433	0.00014738
4	0.00015472	0.00012584	0.00015833	0.00015433	0.00014743

5	0.00015007	0.00012584	0.00015833	0.00015466	0.00014736
6	0.00015448	0.00012741	0.00015833	0.00015466	0.00014701
7	0.00015288	0.00012896	0.00015833	0.00015466	0.00014741
8	0.00015226	0.00012741	0.00015833	0.00015433	0.00014768
9	0.0001507	0.00012361	0.00015833	0.00015466	0.00014773
10	0.00015442	0.00012896	0.00015833	0.00015466	0.00014739
Mean	0.0001524	0.00012739	0.00015833	0.00015461	0.00014745
Std dev.	2.3574E-06	2.4439E-06	0	2.4906E-07	2.0141E-07

Table 4: Calculated Uncertainty Values for Flow 1

	Um, q (m3/s)	Ui, q (m3 /s)	UTotal, q (m3 /s)	Flow Rate (m3/s)*10-4
Venturi	4.08093E-07	4.89689E-06	4.91387E-06	1.632 ± 0.0491
Orifice	3.28429E-07	1.31316E-06	1.35361E-06	1.313 ± 0.0135
Rotameter	2.8571E-20	8.333E-06	8.333E-06	1.667 ± 0.0833
Turbine	1.93243E-07	5.0083E-06	5.01203E-06	1.669 ± 0.0501
Ultrasonic	5.10393E-07	2.40918E-06	2.46265E-06	1.606 ± 0.0264
Bucket				1.6727

Table 5: Calculated Uncertainty Values for Flow 2

	Um, q (m3/s)	Ui, q (m3 /s)	UTotal, q (m3 /s)	Flow Rate (m3/s)*10-4
Venturi	3.81065E-07	4.57187E-06	4.58772E-06	1.524 ± 0.0459
Orifice	3.1857E-07	1.27389E-06	1.31312E-06	1.274 ± 0.0131
Rotameter	0	7.9164E-06	7.9164E-06	1.583 ± 0.0792
Turbine	2.49062E-07	4.63831E-06	4.645E-06	1.546 ± 0.0456
Ultrasonic	2.01407E-07	2.21176E-06	2.22091E-06	1.475 ± 0.0222
Bucket				1.5677

Flowmeter uncertainty analysis involves two key steps: calculating both instrument error and measurement error, and then combining them into a total uncertainty (U_{Total}). This approach applies to Venturi and orifice meters because they have a pressure difference. The manufacturer provides the instrument error ($U_{i,m}$) as a percentage. Measurement error is then determined based on a sample size of 10 and a 95% confidence interval. A t-value of 2.2 is used for all flowmeters. For Venturi and orifice meters, once the ΔP uncertainty is calculated, the flow rate uncertainty can be determined. Flow rate q is a function of ΔP and diameter D_2 . While this experiment assumes perfect diameters, there likely are minor errors that could introduce discrepancies. However, including diameter uncertainty would complicate calculations involving partial derivatives. These calculations require factoring in uncertainties associated with each variable. For simplicity, this experiment assumes ΔP as the sole source of measurement uncertainty. The flow rate measurement uncertainty ($U_{m,q}$) is derived from the partial derivative formula. Finally, the total uncertainty for flow rate ($U_{Total,q}$) in Venturi and orifice meters is calculated by combining the instrument and measurement errors. In contrast, the process for Rotameter, Turbine, and Ultrasonic meters is simpler. Since these meters directly measure flow rate, there's no need for partial derivatives. Therefore, their total uncertainty ($U_{Total,q}$) is calculated using only the instrument error ($U_{i,q}$) and measurement error ($U_{m,q}$).

Figures 1 and 2 show the flow rates for each meter type for flows 1 and 2. The error bars represent the uncertainties associated with each meter's reading. The calculated bucket flow rates for flows 1 and 2 are 0.000167269 m^3/s and 0.000156766 m^3/s , respectively.

Figure 1:

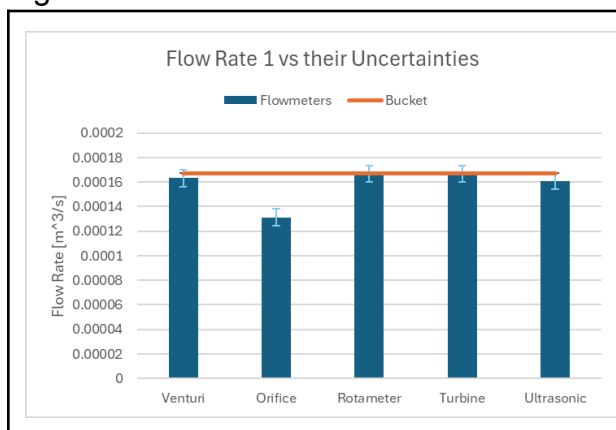
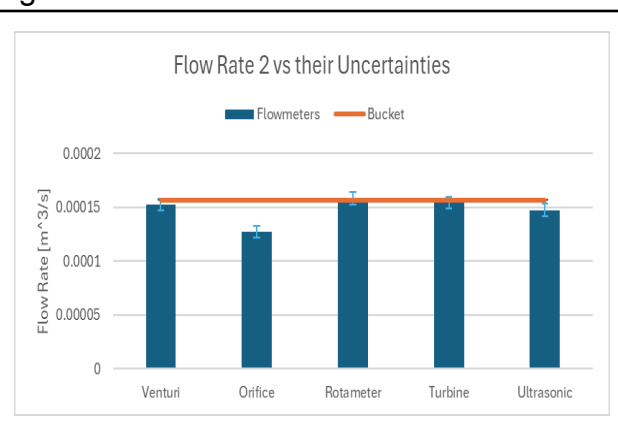


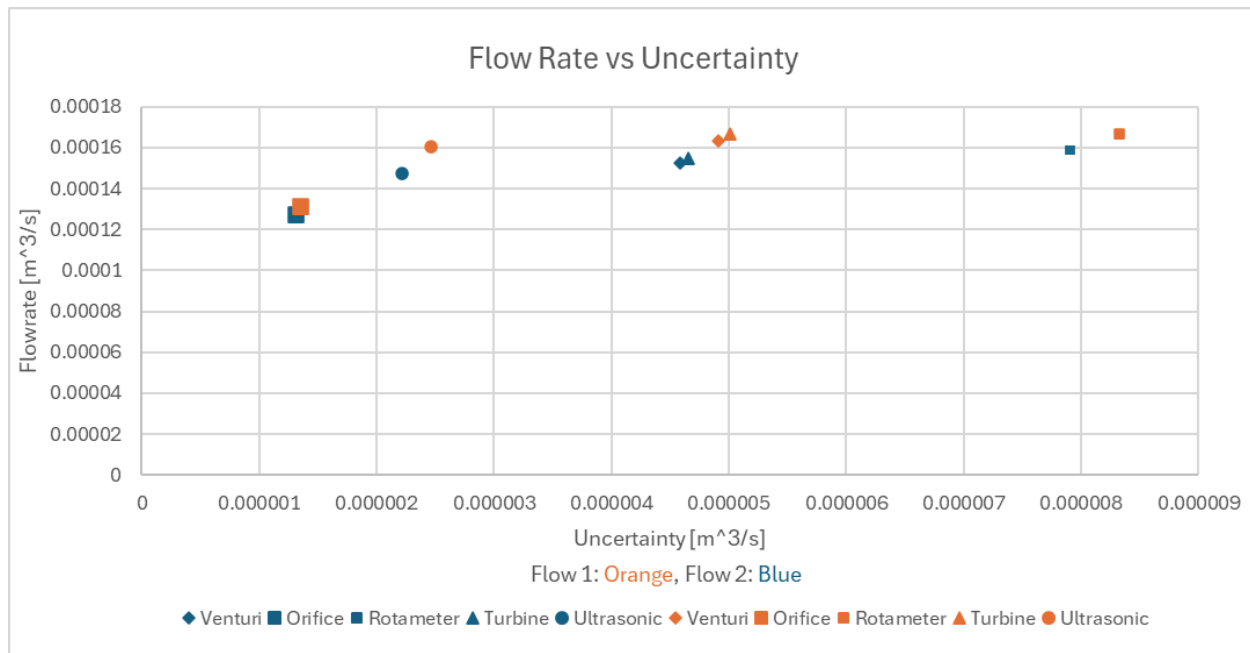
Figure 2:



C) As seen in the figures all meters but the orifice and the ultrasonic in the 2nd flow rate are within the uncertainty. Which is interesting because the orifice meter consistently has the lowest flow rate despite its low error (around $\pm 1\%$). This unexpected outcome

suggests the orifice meter might not be ideal for these measurements. The rotameter readings, while within the uncertainty for both flows, are susceptible to errors due to its analog nature (around $\pm 5\%$ instrumental error). The ultrasonic meter has low error (around $\pm 1.5\%$) similarly to the orifice meter, but just barely falls outside the uncertainty for flow 2. The turbine meter, with a mid-range error (around $\pm 3\%$), performs well in both. The Venturi meter, having the same error as the turbine meter, performs well too. Overall, drawing definitive conclusions is challenging. However, considering both flow rates and their proximity to the expected values, the Rotameter appears to be the most accurate among these meters.

Figure 3:



D) In general, as flow rate increases, the uncertainty in measurement also increases for all flowmeter types. However, orifice and rotameters show a smaller rise in uncertainty compared to other flowmeters as flow rate goes up. Rotameters are the preferred choice for flow rate measurement because they deviate less from the actual flow rate, as shown in Figures 2 and 3. This makes them the most accurate flowmeter among the ones tested.

E) Part C of the experiment employed the "bucket method" to determine flow rate. This involved collecting water in a graduated cylinder for a timed interval. While this method seems straightforward, it lacks an uncertainty analysis. The graduated cylinder's markings might be inaccurate, and human error could lead to misreading the meniscus level. Additionally, human reaction times contribute to potential errors in starting and

stopping the timer. The method doesn't account for flow rate losses within the system due to friction in tubes, bends, and the pump itself. While letting the flow stabilize aims to minimize these discrepancies, it highlights the setup's limitations. The "bucket method" measures flow at the system's end, potentially introducing errors as the water travels through the system accumulating these inaccuracies. Improvements could involve automating the collection process for precise timing and volume measurements. Additionally, measuring flow rate at two opposing points within the system would allow for averaging, effectively canceling out errors due to losses throughout the path.

Conclusion:

This experiment effectively compared the accuracy and effectiveness of different flow rate measuring devices. It highlighted how assumptions in uncertainty calculations can lead to data discrepancies, and how controlled variables can introduce errors. Importantly, the experiment revealed that flow meter effectiveness varies depending on the flow rate scenario. The rotameter's uncertainty, determined by the bucket method, fell within the expected range, but the method itself is imprecise due to timing limitations. Therefore, the ultrasonic meter, with its consistently low uncertainty, emerges as the most reliable option for accurate flow rate measurement.

References:

[1] Carleton University: Department of Mechanical & Aerospace Engineering. "MAAE 2300 Laboratory Manual", 2024

Appendix A: Data/Calculation Tables

Table 6: Pressure and Flow Rate for Flow 1

	Venturi		Orifice	
Point #	ΔP (kPa)	q (m ³ /s)	ΔP (kPa)	q (m ³ /s)
1	2.83	0.000163639	2.21	0.000132612
2	2.73	0.000160722	2.26	0.000134104
3	2.8	0.000162769	2.2	0.000132312
4	2.87	0.000164791	2.15	0.0001308
5	2.82	0.00016335	2.27	0.0001344
6	2.88	0.000165078	2.27	0.0001344
7	2.8	0.000162769	1.98	0.000125522
8	2.8	0.000162769	2.04	0.00012741
9	2.81	0.00016306	2.15	0.0001308
10	2.82	0.00016335	2.15	0.0001308
Mean	2.816	0.00016323	2.168	0.000131316
Std dev.	0.04141927	1.20214E-06	0.097045236	2.9646E-06

Table 7: Pressure and Flow Rate for Flow 2

	Venturi		Orifice	
Point #	ΔP (kPa)	q (m ³ /s)	ΔP (kPa)	q (m ³ /s)
1	2.52	0.000154417	1.99	0.000125839
2	2.3	0.000147522	2.04	0.00012741
3	2.46	0.000152567	2.21	0.000132612
4	2.53	0.000154723	1.99	0.000125839
5	2.38	0.000150066	1.99	0.000125839
6	2.52	0.000154417	2.04	0.00012741
7	2.47	0.000152877	2.09	0.000128962

8	2.45	0.000152257	2.04	0.00012741
9	2.4	0.000150695	1.92	0.000123606
10	2.52	0.000154417	2.09	0.000128962
Mean	2.455	0.000152396	2.04	0.000127389
Std dev.	0.07546154	2.35742E-06	0.078740079	2.44393E-06

Table 8: Uncertainties and Partial Derivatives for Flow 1 and 2

	Flow 1		Flow 2	
	Venturi	Orifice	Venturi	Orifice
U _m , q	4.08093E-07	3.28429E-07	3.81065E-07	3.1857E-07
U _i , q	4.89689E-06	1.31316E-06	4.57187E-06	1.27389E-06
U _{Total} , q	4.91387E-06	1.35361E-06	4.58772E-06	1.31312E-06
U Δ P	14.08029486	10.842102	12.2761226	10.201471
U _m , P	0.091122384	0.2134995	0.166015394	0.1732282
U _i , P	14.08	10.84	12.275	10.2
$\delta q / \delta \Delta P$	2.89833E-08	3.0292E-08	3.10412E-08	3.12279E-08
q _{Measured} (m ³ /s)	0.000167269	0.000167269	0.000156766	0.000156766

Table 9: Uncertainties for Flow 1

	Rotameter	Turbine	Ultrasonic
U Δ P	50	50.08501	48.18555
U _m , P	0	0.025509	0.067375
U _i , P	50	50.085	48.1855
U _m , q	2.8571E-20	1.93243E-07	5.10393E-07
U _i , q	8.333E-06	5.0083E-06	2.40918E-06
U _{Total} , q	8.333E-06	5.01203E-06	2.46265E-06

Table 10: Uncertainties for Flow 2

	Rotameter	Turbine	Ultrasonic
$U_{\Delta P}$	47.5	46.38501	44.23701
U_m, P	0	0.032878	0.026587
U_i, P	47.5	46.385	44.237
U_m, q	0	2.49062E-07	2.01407E-07
U_i, q	7.9164E-06	4.63831E-06	2.21176E-06
U_{Total}, q	7.9164E-06	4.645E-06	2.22091E-06

Appendix B Sample Calculations:

Means/Standard deviation done for both Pressure P and Flow Rate q
(Only Pressure sample calculations shown below)

$$\text{Mean} = \Delta P = (2.83 + 2.73 + 2.8 + 2.87 + 2.82 + 2.88 + 2.8 + 2.8 + 2.81 + 2.82)/10 \\ = 2.816 \text{ kPa}$$

$$\text{Std. dev} = s = (((2.83 - 2.816)^2 + \dots + (2.82 - 2.816)^2)/10)^{1/2} \\ = 0.041419266$$

$$U_m, \Delta P = 2.2 * (0.041419266) = 0.091122384$$

$$U_i, \Delta P = 0.005 * 2.816 * 1000 = 14.08$$

$$U_i, q = 0.03 * 0.00016323 = 4.89689\text{E-}06$$

$$U\Delta P = (14.08^2 + 0.091122384^2)^{1/2} = 14.08029486$$

$$D1 = 29.455 \text{ mm}$$

$$D2 = 9.525 \text{ mm}$$

$$d = 29.455 / 9.525 = 0.323375$$

$$\delta q / \delta \Delta P = (3.14 * 0.96 * (9.525/1000)^2) / \\ (4 * \text{SQRT}(2000) * \text{SQRT}(2.816 * 1000) * \text{SQRT}(1 - 0.323374639^4)) \\ = 2.89833\text{E-}08$$

$$U_m, q = ((2.89833\text{E-}08 * 14.08)^2)^{1/2} \\ = 4.08093\text{E-}07 \text{ m}^3/\text{s}$$

$$U_{\text{Total}}, q = ((3.82364 * 10^{-7})^2 + (4.58794 * 10^{-6})^2)^{1/2} \\ = 4.91387\text{E-}06 \text{ m}^3/\text{s}$$

$$q_{\text{Actual}} \\ = 0.96 * (3.14/4) * ((9.525/1000)^2) * (\text{SQRT}((2 * (B3 * 1000)) / (1000 * (1 - (0.323375^4))))) \\ = 0.000153803 \text{ [m}^3/\text{s]}$$

$$q_{\text{Measured}} = ((925/1000000)/5.53) = 0.000167269 \text{ m}^3/\text{s}$$

$$q_{\text{Exp.}} = (0.00016323 \pm 0.00000491387) \text{ m}^3/\text{s}$$